



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

Department of Computer Science
Faculty of Computing

SECP3213 – BUSINESS INTELLIGENCE

(Project Proposal)

Behind the Delay: A Business Intelligence Analysis of Airline Performance and Passenger Satisfaction

Programme	: Bachelor of Computer Science (Data Engineering)
Subject Code	: SECP3213
Subject Name	: Business Intelligence
Session-Sem	: 2025/2026-02

Prepared by : 1) Muhammad Afiq Danish bin Mohd Hazni (A23CS0118)
2) Muhammad Adam bin Razali (A23CS0116)

Section : 02

Lecturer : Dr. Noorfa Haszlinna binti Mustaffa

Table of Contents

1.0 Problem Background.....	2
1.1 Business Context.....	2
1.2 Importance of the Problem.....	2
1.3 Expected Impact.....	3
2.0 Objectives.....	4
3.0 Datasets Description.....	4
3.1 Dataset 1: Flight Delay & Cancellation Data.....	4
3.2 Dataset 2: Airline Passenger Satisfaction.....	5
3.3 Dataset 3: Airline Delay Causes.....	6
4.0 Proposed BI Approach.....	7
4.1 Tool Usage - Alteryx Designer and Power BI.....	7
4.2 Data Integration, Data Cleaning and Data Transformation.....	7
4.3 Visual Analysis.....	8

1.0 Problem Background

1.1 Business Context

The commercial aviation industry is one of the most operationally complex sectors in the world involving the coordination of thousands of flights, airlines, airports, air traffic controllers and ground crews on a daily basis. Despite advancements in technology and operational management, flight delays remain one of the most consistent and costly challenges facing the industry. The aviation industry continues to struggle with the financial burden of flight delays which collectively cost airlines billions of dollars annually in operational expenses such as additional fuel consumption, crew overtime and aircraft repositioning costs.

Flight delays arise from a wide range of causes including adverse weather conditions, air traffic congestion managed by the National Aviation System (NAS), late-arriving aircraft, airline carrier issues such as maintenance or crew unavailability and security issues. The natural causes of delay makes it particularly challenging for airlines and airport authorities to isolate root causes and implement accurate corrective measures. Furthermore, a single delayed flight can trigger a chain reaction of disruptions across multiple routes and schedules which worsen the overall operational and financial damage.

Beyond the operational perspectives, flight delays carry significant implications for customer experience and brand loyalty. Passengers who experience delays particularly frequent or business travellers often respond with lower satisfaction ratings, reduced brand loyalty and an increased likelihood of switching to competing and rival airlines. Understanding the relationship between delay patterns and passenger satisfaction is therefore not only an operational concern but also a critical strategic and commercial priority for airline management.

1.2 Importance of the Problem

The importance of addressing flight delay analysis through a Business Intelligence (BI) approach lies in the industry's need for data-driven decision making. Traditionally, airlines tend to respond to delays only after they occur rather than taking preventive measures based on data and insights. By integrating data from multiple sources including flight performance records, delay cause classifications and passenger satisfaction surveys, a BI solution can provide a comprehensive and holistic view of airline performance that supports more informed and timely decisions.

Operations managers can make better decisions by understanding which delay causes occurs most frequently and which ones can be prevented, allowing them to invest in the right improvements. For customer experience teams, understanding how delay severity and type affect satisfaction scores enables better passenger communication strategies and service recovery frameworks. For executive leadership, comparing their on-time performance against other airlines helps them understand where they stand in the industry and make better strategic decisions.

Without a structured BI framework, this analysis would remain fragmented across siloed departmental reports which make it difficult to derive cross-functional insights. This project addresses the gap by building an integrated BI solution that connects operational delay data with customer satisfaction outcomes.

1.3 Expected Impact

The expected outcomes and impact of this project include the following:

1. **Improved Operational Decision-Making:** Airlines and airport authorities will be able to identify the most significant and controllable causes of delays to enable targeted operational improvements and resource allocation strategies.
2. **Enhanced Passenger Experience:** By uncovering the relationship between delay patterns and passenger satisfaction, airlines can develop more effective communication strategies and service recovery protocols to mitigate customer dissatisfaction.
3. **Data-Driven Policy Recommendations:** Findings from the combined datasets will help airline management make informed decisions including how to schedule flights more efficiently, manage the crews better and plan for weather disruptions.
4. **Foundation for Predictive Analytics:** The cleaned and integrated dataset produced through this project will serve as a foundation for future predictive modelling efforts such as forecasting delay possibilities based on historical patterns.

2.0 Objectives

This project is guided to achieve these following objectives:

1. Preparing flight delay datasets including flight delay records, delay causes and customer satisfaction surveys using Alteryx Designer for preparation of the data and Power BI for analytical visualizations.
2. To identify the primary causes and patterns of flight delays across different airlines, airports, time periods, and route segments through data transformation and exploratory analysis.
3. To analyse the relationship between specific delay categories such as carrier, weather, NAS, security and late aircraft, which quantify the customer satisfaction.
4. Designing an interactive dashboard via a data-driven insight as well as a storyboard supporting the fact for airline management.
5. To formulate actionable business recommendations based on the BI findings that can guide airlines and airports in reducing avoidable delays and improving overall service quality.

3.0 Datasets Description

This project integrates three publicly available datasets which all sourced from Kaggle. Each dataset contributes a different analytical dimension to the business problem, collectively enabling a comprehensive examination of airline delay patterns, root causes and their impact on passenger satisfaction.

3.1 Dataset 1: Flight Delay & Cancellation Data

Attribute	Details
Source	Kaggle
URL	https://www.kaggle.com/datasets/nalisha/flight-delay-and-cancellation-data-1-million-2024
Data Type	Structured
Context	Over 1 million US airline performance records for the year 2024

Key Features:

- *fl_date* - Date of the flight
- *origin* - Origin airport code
- *origin_city_name* - Full city name of the departure airport
- *cancelled* - Binary indicator of whether a flight was cancelled
- *air_time* - Actual time spent in the air in minutes
- *distance* - Flight distance in miles
- *weather_delay* - Delay duration in minutes attributed to weather conditions
- *late_aircraft_delay* - Delay duration in minutes caused by a late-arriving previous aircraft

This dataset serves as the core operational dataset for the project. It provides over one million flight records covering US domestic routes throughout 2024, making it a large-scale and current source of airline performance data. Key metrics such as cancellation status, weather-related delays, and late aircraft delays form the foundation of the operational analysis, enabling the identification of delay trends across different airports, routes, and time periods.

3.2 Dataset 2: Airline Passenger Satisfaction

Attribute	Details
Source	Kaggle
URL	https://www.kaggle.com/datasets/teejmahal20/airline-passenger-satisfaction
Data Type	Structured
Context	Survey responses collected from airline passengers across multiple airlines

Key Features:

- *Gender* - Gender of the passenger (Male/Female)
- *Customer Type* - Indicates whether the passenger is a loyal or disloyal customer
- *Age* - Age of the passenger
- *Type of Travel* - Purpose of travel (Business/Personal)
- *Class* - Travel class (Business, Economy, Economy Plus)
- *Flight Distance* - Distance of the journey in miles
- *Departure Delay in Minutes* - Number of minutes the flight was delayed at departure
- *Arrival Delay in Minutes* - Number of minutes the flight was delayed upon arrival
- *On-board Service* - Passenger satisfaction rating for onboard service quality (1-5)
- *Satisfaction* - Overall satisfaction outcome (Satisfied/Neutral or Dissatisfied)

This dataset provides the customer impact dimension of the analysis. By linking passenger satisfaction ratings with delay experience and travel characteristics, it enables the project to measure the real-world business consequences of operational delays on customer experience. The inclusion of loyalty status, travel class and demographic variables further allows for segmented analysis to understand which passenger groups are most sensitive to delays which provide valuable insights for airline customer experience strategy.

3.3 Dataset 3: Airline Delay Causes

Attribute	Details
Source	Kaggle
URL	https://www.kaggle.com/datasets/abdalrhmantwfi/airline-delay-causes
Data Type	Structured
Context	Natural causes of airline delay across multiple years

Key Features:

- *year* - The year of the collected data
- *month* - The month of the collected data
- *carrier* - Airline carrier code
- *carrier_name* - Airline carrier full name
- *airport* - Unique airport code
- *airport_name* - Full name and location of the airport
- *arr_flights* - Total number of arriving flights during the period
- *arr_dell5* - Number of flights delayed by 15 minutes or more
- *carrier_ct* - Number of delay minutes attributed to airline-controllable factors such as maintenance issues and crew performance
- *weather_ct* - Number of delay minutes attributed to weather conditions
- *nas_ct* - Number of delay minutes attributed to the National Airspace System including air traffic control and airport operations
- *security_ct* - Number of delay minutes attributed to security-related issues
- *late_aircraft_ct* - Number of delay minutes caused by late-arriving aircraft from a previous flight
- *arr_cancelled* - Number of cancelled flights

This dataset provides the diagnostic layer of the analysis. By breaking down delays into five distinct categories which are carrier, weather, NAS, security and late aircraft, it enables the identification of the most significant contributors to delays at both airline and airport level. Crucially, it allows the project to distinguish between controllable delays and uncontrollable delays which carry different implications for operational strategy and business decision-making.

4.0 Proposed BI Approach

4.1 Tool Usage - Alteryx Designer and Power BI

Alteryx Designer will be our main data preparation tool for this project paired with Power BI as a visualization tool that feeds on the processed data from the preparation tool. All 3 datasets will be ingested, cleaned, transformed, and integrated within a single Alteryx Workflow of (.yxmd). Power BI will then connect directly with the output data from the workflow to later produced dashboards with storytelling.

4.2 Data Integration, Data Cleaning and Data Transformation

Data Integration

1. Summarization tool - Dataset 3 is aggregated to compute the average delay cause for each airport and month across all airlines and years. This can later be matched to the flight records from dataset 1.
2. Each flight record in Dataset 1 will then be matched with the aggregated data earlier to enrich it about the typical causes that delays the airport flight. The new records will include historical data of the shared delays caused by the airline which are weather, air traffic control, security, and late incoming aircraft.
3. For the Dataset 2 related to customer satisfaction. It will be separately managed on its own workflow, since it cannot be linked to other datasets via any identifiers.

Data Cleaning

1. Rows with missing values or incorrect values will be imputed or complimented from the datasets.
2. Duplicate flight records will be identified and removed based on a combination of flight date, origin, destination, and scheduled flight duration.
3. All date fields will be set to the correct date format, delay times will be treated as numbers, and airline and airport codes will be standardized to a consistent format.
4. Where a flight has no cancellation recorded, it will be treated as not cancelled. Where delay cause totals are missing after the summarisation step, they will be set to zero and flagged so that this is visible in the analysis.

Data Transformation

1. Formulation tool - Each flight will be assigned a delay severity label based on the available delay minutes, since the delays recorded are disparate and not combined to be analytical-ready.
2. The flight records will be summarised to airport level for airport-based performance dashboard and the Airline Delay Causes data will be summarised to airline level for the carrier comparison dashboard.
3. In the satisfaction dataset, a matching delay severity label will be added based on each passenger's recorded departure delay, so that satisfaction outcomes can be compared across delay groups.

4.3 Visual Analysis

For the dashboard, we will be including at least 5 of these charts with said justifications as explained below:

Chart Type	Justification
Line Chart	Tracking the average delay times by the change of month and year based on the Airline Delay Causes dataset
Stacked Bar Chart	Shows the number of airline's total delay by their specific causes such as weather, airline's own operations, or air traffic control. Using side-by-side bars making it easier to compare both the total amount of delay with the airline names.
Ranked Bar Chart	Ranks each services from different airline on how satisfied their passengers are based on the scoring. These could bring up insights about what actually influencing the passenger onboarding experiences.
Heat Map	Shows a delay frequency for each of the unique origin airport using a color-coded grid. This ensures easiness in identifying problematic airport immediately.
Scatter Plot	Plot departure delay length against passenger satisfaction to derive relationship between the two. Points representing longer

	delays are expected to cluster in the dissatisfied zone, helping to identify the delay length at which satisfaction drops most sharply. This uses the passenger satisfaction dataset, which contains both delay figures and satisfaction ratings.
KPI Summary Cards	Display the key figures for the overall on-time rate, average delay in minutes, cancellation rates, and most common delay cause as a quickshot of attention for customer to understand the charts in more detail